Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/27/2006
PI Name:	Hall, Eric J Ph.D., D.Sc.		
Project Title:	Individual Genetic Susceptibility		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ejh1@columbia.edu	**Fax:**	FY 212-305-3229
PI Organization Type:	UNIVERSITY	**Phone:**	212-305-5660
Organization Name:	Columbia University		
PI Address 1:	630 West 168th Street		
PI Address 2:	Center for Radiological Research		
PI Web Page:	http://cpmcnet.columbia.edu/		
City:	New York	**State:**	NY
Zip Code:	10032	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	05/19/2003	**End Date:**	05/31/2007
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: end date changed to 5/31/2007 per J. Dardano (5/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Brenner, David J (Columbia University) Smilenov, Lubomir (Columbia University)		
Grant/Contract No.:	NAG 9-1519		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The working hypothesis of this proposal was that cells from heterozygous ATM knockout mice are more sensitive to radiation than cells from wild-type mice. Two endpoints related to radiosensitivity were tested. Mouse embryo cells from wild-type and knockout animals were irradiated and the incidence of oncogenic transformation scored; this is an example of a stochastic effect. Radiation induced cataracts were studied in adult wild-type and knockout mice; this represents an important deterministic effect. Oncogenic Transformation Mice heterozygous for the ATM gene were mated, mid-term pregnant animals sacrificed and cell cultures prepared from individual embryos. The cells were genotyped as wild-type, homozygous or heterozygous for the ATM gene. All three cell types (wild-type, homozygous and heterozygous for the ATM gene) were exposed to gamma-rays. There was a statistically significant increase in the incidence of oncogenic transformation in cells from AT heterozygotes compared with wild-type cells from litter-mates.		

	Cataractogenesis Wild-type mice, as well as mice heterozygous for the ATM gene were irradiated with graded doses of x-rays and observed regularly for cataract development using both subjective (slit-lamp) biomicroscopy and quantitative objective (Scheimpflug Imaging) measures. Preliminary data indicate that cataracts develop earlier and reach a higher grade in AT heterozygous mice, compared with wild-type animals.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The dose limits proposed for astronauts in space flight are different from those on the ground, though the basic philosophy of limiting the life-time risk of stochastic effects to 3% is preserved. There are several reasons for the differences, including the limited number of individuals exposed, the practicality of shielding and the presence of more urgent risks. Nevertheless the purpose of radiation protection remains to prevent deterministic effects of clinical significance and limit stochastic effects to acceptable levels. The levels proposed are based on the assumption that the human population is uniform in its radiosensitivity. Using two endpoints, namely cataracts as an important and relevant deterministic effect and oncogenic transformation in cultured embyro cells as a surrogate for carcinogenesis, the most relevant stochastic effect, we have shown that mice heterozygous for the atm gene are more radiosensitive than wild type animals. 1 to 3% of the US population are AT heterozygotes, and it would not be prudent to subject such individuals to substantial doses of radiation.
Task Progress:	In the last several years, heterozygocity leading to haploinsufficiency for proteins involved in DNA damage processing was shown to play a role in genomic instability and tumorigenesis after DNA damage is induced. It was also shown that heterozygocity is a factor in tumorigenesis that depends of the genetic background. Genes that confer radiosensitivity now include rad9, ATM, and BRCA1. It is also evident that double heterozygotes are even more radiosensitive. Our ongoing experiments with gamma-rays and with heavy ions at Brookhaven National Laboratory clearly demonstrate that animals haploinsufficient for one or a more genes involved in DNA repair are more sensitive to high energy heavy ions than wild-type animals. The effect of double heterozygocity for ATM/Rad9 and ATM and BRCA1 genes is even stronger than the effect of single heterozygocity. These data strongly indicate that individual radiation sensitivity is genetically based and possibly could be determined by genetic analysis. This addresses CPR question 10.12 concerning inter-individual diversity. It is already evident that the human population is not homogeneous for radiosensitivity. The pattern of response for the increase of apoptosis with dose is qualitatively similar for gamma-rays and high energy heavy ions, but quantitatively different, i.e. high energy heavy ions are more effective for a given dose. However, the biological effects produced by the two radiations are similar for gamma-rays and high energy heavy ions, but quantitatively different, i.e. high energy heavy ions are more effective for a given dose. However, the biological effects produced by the two radiations are similar. There is no unique effect associated with the HZE. This addresses CPR question 10.05 concerning whether there are unique biological effects associated with HZEs.
Bibliography Type:	Description: (Last Updated: 10/26/2023)
Articles in Peer-reviewed Journals	Smilenov LB. "Tumor development: haploinsufficiency and local network assembly." Cancer Lett. Epub 2005 Oct 11. PMID: 16223564 , Oct-2005
Articles in Peer-reviewed Journals	Sokolov MV, Smilenov LB, Hall EJ, Panyutin IG, Bonner WM, Sedelnikova OA. "Ionizing radiation induces DNA double-strand breaks in bystander primary human fibroblasts." Oncogene. 2005 Nov 10;24(49):7257-65. PMID: 16170376 , Nov-2005
Articles in Peer-reviewed Journals	Smilenov LB, Lieberman HB, Mitchell SA, Baker RA, Hopkins KM, Hall EJ. "Combined haploinsufficiency for ATM and RAD9 as a factor in cell transformation, apoptosis, and DNA lesion repair dynamics." Cancer Res. 2005 Feb 1;65(3):933-8. PMID: 15705893 , Feb-2005
Articles in Peer-reviewed Journals	Hall EJ, Brenner DJ, Worgul B, Smilenov L. "Genetic susceptibility to radiation." Adv Space Res. 2005;35(2):249-53. , Feb-2005
Articles in Peer-reviewed Journals	Worgul BV, Smilenov L, Brenner DJ, Vazquez M, Hall EJ. "Mice heterozygous for the ATM gene are more sensitive to both X-ray and heavy ion exposure than are wildtypes." Adv Space Res. 2005;35(2):254-9. , Feb-2005